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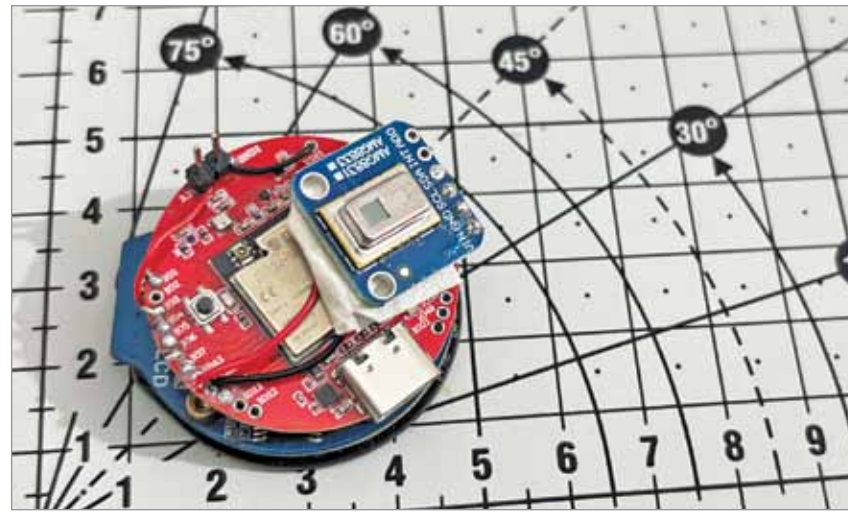


Fig. 5: Prototype of thermal camera assembled and soldered, along with a size scale for comparison

```
void loop() {
  // Move the heat blob
  blobX += velX;
  blobY += velY;

  // Bounce
  if (blobX < 5 || blobX > GRID - 6) velX = -velX;
  if (blobY < 5 || blobY > GRID - 6) velY = -velY;

  // Slowly cool the whole image
  for (int y = 0; y < GRID; y++) {
    for (int x = 0; x < GRID; x++) {
      heat[y][x] = heat[y][x] * 0.88 + 24.0 * 0.12;
    }
  }

  // Add a large realistic heat blob (like a finger)
  addHeatBlob(blobX, blobY, 34.5, 5.5);   // main hot core
```

Fig. 6: Thermal camera code snippet for mapping thermal data to display colours

detection is 20°C to 38°C.

4. Incorrect temperature reading. Ensure that the sensor is uncovered and properly facing the target. The centre temperature is calculated from the middle pixels of the 8 × 8 array.

5. Device resets or does not power on. Check the battery voltage and VCC/GND connections. Poor power connections can cause random resets.

Once the thermal image appears correctly and responds to heat sources, the camera can be used for basic applications such as PCB fault detection, temperature monitoring, and thermal analysis. EFY

TO DOWNLOAD SOURCE CODE AND RESOURCES

Scan this QR code or go to https://sourcecode.electronicsforu.com to access and download source code and other resources for this project

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